PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Food Waste Prediction Using Machine Learning Models

submitted in partial fulfilment of the requirements for the award of degree of

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In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **“"Food Waste Prediction Using Machine Learning Models”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of <**Gurpreet Singh**>. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

A huge amount of food gets wasted every year all over the world. Around one-third of all food produced never gets eaten, and in India alone, households waste about 78 million tonnes of food every year. This is a big problem because so many people still go hungry, and wasted food also harms the environment.

In this project, I have studied how machine learning can be used to predict food waste before it actually happens. If we can predict how much food will be wasted, restaurants, shops, and households can plan better and reduce waste. I reviewed several machine learning methods like Random Forest, LSTM (a type of neural network), XGBoost, and CNNs (for image-based spoilage detection), and compared how well they work in different situations.

The results show that these models can be quite accurate, some achieving over 90% accuracy, and can help cut food waste by 14% to 52% in real settings. However, there are also some challenges like getting good data, making models work in different places, and making them easy enough for small businesses to use.

Keywords: food waste, machine learning, Random Forest, LSTM, prediction, sustainability

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1. Introduction

Food waste is a serious problem all over the world. According to the United Nations, about one-third of all food produced globally gets wasted every year. In India, this problem is even bigger because a lot of fresh food spoils before it even reaches the consumer. According to estimates, Indian households waste around 78 million tonnes of food every year, which comes out to roughly 50 kilograms per person annually.

This is not just a waste of food — it also wastes the water, land, and energy that were used to grow that food. On top of that, rotting food releases harmful gases like methane into the atmosphere, which contributes to climate change. The United Nations has set a goal (SDG 12.3) to cut food waste in half by 2030, but reaching that target is going to require smarter tools and better planning.

1.1 Why Machine Learning?

In the past, people tried to manage food waste by manually tracking how much was thrown away or by using rough guesses based on past experience. These methods are not very accurate and are hard to scale. Machine learning is a better approach because it can find hidden patterns in large amounts of data and make accurate predictions about future waste.

For example, a restaurant can use a machine learning model to predict how much food it will need on a particular day based on factors like the day of the week, the weather, upcoming holidays, and past sales. This way, they can order just the right amount and avoid throwing food away at the end of the day.

1.2 Objectives of the Project

- To understand and compare different machine learning methods used for food waste prediction.
- To see how well these methods work in different situations like restaurants, retail stores, and homes.
- To identify the main challenges in actually using these systems in real life.
- To suggest improvements that could make these systems more useful, especially in India.

1.3 What This Report Covers

This report looks at research done between 2023 and 2026 on using machine learning for food waste prediction. It covers different types of models — from simple decision trees to complex neural networks, and compares them based on how accurate they are and how practical they are to use in the real world.

2. Methodology

For this project, I did a literature review — meaning I read and summarised research papers and reports published on the topic of food waste prediction using machine learning. I looked at papers from databases like IEEE Xplore, Google Scholar, and Scopus. My main search keywords were things like 'food waste machine learning', 'demand forecasting food', 'IoT food waste', and 'spoilage detection deep learning'.

I focused on papers from 2023 to 2026 that actually tested their models on real data and reported numbers like accuracy or waste reduction. This helped me compare the methods fairly.

2.1 Data Used in These Studies

Different studies used different kinds of data. Some common sources were:

- Public datasets from Kaggle, FAO, and UNEP.

- Sales records from restaurants and stores.
- Data from smart bins and IoT sensors that measure weight and fill levels.
- Survey data collected from households about their food habits.
- Weather data and holiday calendars to understand demand patterns.

2.2 Important Features

The features (inputs) used in these models can be grouped into four categories:

- Time-related: day of the week, time of day, season, holidays.
- Environmental: weather conditions, storage temperature.
- Contextual: type of venue, customer demographics.
- Operational: current stock levels, portion sizes, batch sizes.

2.3 Data Challenges

Working with food waste data is not straightforward. Some common problems I noticed across the papers I reviewed were: missing data (especially from sensor failures), very unequal class distributions (high-waste events are rare), noisy or inconsistent records, and privacy concerns when collecting data from homes. Most studies dealt with this by cleaning the data, filling in missing values, and using techniques like oversampling to balance the classes.

3. Tools and Technologies

The studies I reviewed used a variety of tools and technologies to build and test their models. Here is a summary of the most commonly used ones.

3.1 Programming Language

- Python — used in almost all the studies because it has great libraries for machine learning and data analysis.

3.2 Machine Learning Libraries

- Scikit-learn — for models like Random Forest, SVM, Decision Trees, and KNN.
- XGBoost / LightGBM — for gradient boosting models that work well on tabular data.
- TensorFlow / Keras and PyTorch — for building deep learning models like LSTM and CNN.

3.3 Data and Visualisation Tools

- Pandas and NumPy — for handling and processing data.
- Matplotlib and Seaborn — for plotting charts and graphs.
- SHAP — for understanding which features are most important in a model's predictions.

3.4 IoT Hardware (used in smart bin studies)

- Weight sensors — to measure how much food is being thrown away.
- Ultrasonic sensors — to check how full a bin is.
- Temperature sensors — to track storage conditions.

3.5 Deployment and Cloud Tools

- AWS and Google Cloud — for storing data and running models online.
- Docker and REST APIs — for packaging and deploying models in applications.
- Jupyter Notebook — for experimenting with code and visualising results.

4. Implementation

Based on the papers I reviewed, a typical food waste prediction system follows a similar pipeline. Here is how it works, step by step.

Step 1 – Collect Data

First, you need to gather the data. This could be sales records from a restaurant, sensor readings from smart bins, or survey responses from households. You also bring in external data like weather forecasts and holiday calendars to make the predictions more accurate.

Step 2 – Clean the Data

Raw data is usually messy. You need to remove outliers, fill in missing values, and convert things like 'Monday' or 'Winter' into numbers that the model can understand. You also need to scale the features so that large numbers don't dominate the model.

Step 3 – Pick the Right Features

Not all features are equally useful. Techniques like correlation analysis help identify which inputs actually matter. The data is then split into three parts: 70% for training, 15% for validation, and 15% for testing. Since this is time-series data, the split must follow the order of time — you can't test on data from the past if you trained on the future.

Step 4 – Train the Model

Several models are tried and compared — for example, Random Forest, XGBoost, and LSTM. Their settings (called hyperparameters) are tuned using the validation set to get the best possible performance.

Step 5 – Evaluate the Results

The trained model is tested on the held-out test set. The performance is measured using metrics like MAE (how far off the predictions are on average), RMSE, and accuracy. The model is also compared to a simple baseline like 'predict yesterday's value'.

Step 6 – Use the Model

Once the model is working well, it gets connected to a dashboard or app. For example, it might send a daily recommendation to a restaurant kitchen saying 'reduce tomorrow's order by 20%'. The model keeps learning from new data over time to stay accurate.

4.1 Which Model to Use When?

Random Forest and XGBoost work best when you have a mix of different types of features (like numbers, categories, and dates). LSTM is better when the patterns span several days or weeks in sequence, like weekly demand cycles. CNNs are the right choice when you want to detect spoilage from food photos. Reinforcement learning is useful for smart inventory management where decisions need to adapt in real time.

5. Results and Findings

Here I summarise the main results from the papers I reviewed, grouped by where the model was being used.

5.1 Restaurants and Catering

Rodrigues et al. (2024) tested several models for predicting daily food demand in catering services. Random Forest and LSTM performed the best, and when kitchens used these predictions to plan how much food to prepare, waste went down by 14% to 52%. That is a really significant improvement compared to just guessing based on past averages.

5.2 Smart Bins and IoT Systems

Nemade et al. (2023) built a system using weight and fill-level sensors connected to a cloud-based ML model. The model could identify the type of waste and predict when the bin would overflow with about 93% accuracy. This kind of system can be really useful in large canteens or food courts.

5.3 Food Factories and Processing Plants

Machireddy (2024) looked at ML for industrial food production. Models used for forecasting supply and demand in factories achieved around 92% precision in predicting waste volumes, which helped factories better align production with actual need.

5.4 Households

Some studies used survey data to understand waste behaviour in homes. Singh et al. (2025) reported classification accuracies as high as 99% using ensemble methods. However, this should be taken with some caution because the data came from self-reported surveys, and people do not always accurately report how much food they waste.

5.5 Detecting Spoilage from Images and Sensors

Fotovvatikhah et al. (2025) used CNNs along with gas sensors to detect food spoilage early. In real store and restaurant trials, this led to waste reductions of about 15% to 30% by helping staff know which items to use or discount first.

Table 1 below compares the models and their results:

Model	Where It Was Used	Main Metric	Result
Random Forest	Restaurants / retail	Waste reduction	14–52%
LSTM	Time-series food demand	Prediction accuracy	>90%
XGBoost	Food factories	Precision	~92%
IoT + ML	Smart bins	Classification accuracy	~93%
CNN + sensors	Spoilage detection	Waste reduction	15–30%
Ensemble (surveys)	Household behaviour	Classification accuracy	Up to 99%

Table 1: Comparison of ML Models for Food Waste Prediction

6. Conclusion and Future Scope

6.1 Conclusion

From my research, it is clear that machine learning can be a very useful tool for reducing food waste. Models like Random Forest and LSTM have shown strong results in real-world tests, helping to cut waste by significant amounts. The technology is already being used in some restaurants and stores, and the results are promising.

That said, there are still some challenges. Getting good quality data is not easy, especially in household settings or in smaller cities. Deep learning models are very accurate but hard to explain, which makes it difficult to convince business owners to trust them. And many small businesses simply do not have the budget or infrastructure to set up these systems.

In a country like India, where food habits vary a lot across regions and the food system is quite different from Western countries, we cannot just apply the same models that work abroad. We need models that are built specifically for our context — taking into account local festivals, monsoon seasons, and the fact that a lot of food supply chains are still quite informal.

6.2 Future Scope

There are several things that could be done to improve this field in the future:

- Federated Learning: Instead of sending everyone's data to one central server, models can be trained locally on each device (like a restaurant's computer or a home smart device) and only share the learnings. This protects privacy while still improving the model.
- Explainable AI: Making models easier to understand will help business owners and policy makers trust and use them. Tools like SHAP can help explain why a model is making a certain prediction.
- India-Specific Models: More research is needed on building models that are trained on Indian data and account for local seasons, festivals, and eating habits.
- Combining Prediction with Action: Instead of just predicting waste, future systems could automatically take action, like adjusting orders or sending surplus food to NGOs — without needing human intervention.
- More Household Research: Most studies focus on large institutions. We need more research on how to help regular households reduce their food waste using simple, affordable tools.

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APPENDICES

Appendix A: List of Abbreviations

Abbreviation	Full Form
ML	Machine Learning
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
CNN	Convolutional Neural Network
IoT	Internet of Things
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
SHAP	SHapley Additive exPlanations
SDG	Sustainable Development Goal
FAO	Food and Agriculture Organization
UNEP	United Nations Environment Programme

Appendix B: Sample Dataset

The table below shows what a typical dataset for this kind of project might look like:

Feature	Type	Example
date	Date	2024-03-15
day_of_week	Category	Friday
is_holiday	Yes/No	No
season	Category	Spring
guest_count	Number	320
temperature_C	Number	22.5
waste_kg	Number (target)	18.3